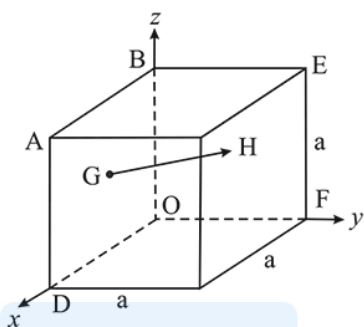


JEE Mains 2019 Chapter wise Question Bank

Mathematics in Physics - Questions

Q1

In the cube of side 'a' shown in the figure, the vector from the central point of the face ABOD to the central point of the face BEFO will be:



- (1) $\frac{1}{2}a(\hat{k} - \hat{i})$ (2) $\frac{1}{2}a(\hat{i} - \hat{k})$
 (3) $\frac{1}{2}a(\hat{j} - \hat{i})$ (4) $\frac{1}{2}a(\hat{j} - \hat{k})$

10 Jan Morning

Q2

Two forces P and Q, of magnitude 2F and 3F, respectively, are at an angle θ with each other. If the force Q is doubled, then their resultant also gets doubled. Then, the angle θ is:

- (1) 120° (2) 60°
 (3) 90° (4) 30°

10 Jan Evening

Q3

Two vectors $\vec{A}$ and $\vec{B}$ have equal magnitudes. The magnitude of $(\vec{A} + \vec{B})$ is 'n' times the magnitude of

$(\vec{A} - \vec{B})$. The angle between $\vec{A}$ and $\vec{B}$ is:

- (1) $\cos^{-1} \left[\frac{n^2 - 1}{n^2 + 1} \right]$ (2) $\cos^{-1} \left[\frac{n - 1}{n + 1} \right]$
 (3) $\sin^{-1} \left[\frac{n^2 - 1}{n^2 + 1} \right]$ (4) $\sin^{-1} \left[\frac{n - 1}{n + 1} \right]$

10 Jan Evening

Q4

Let $|\vec{A}_1| = 3$, $|\vec{A}_2| = 5$ and $|\vec{A}_1 + \vec{A}_2| = 5$. The value of $(2\vec{A}_1 + 3\vec{A}_2) \cdot (3\vec{A}_1 - 2\vec{A}_2)$ is:

- (1) -106.5 (2) -99.5 (3) -112.5 (4) -118.5

8 April Evening

Q5

The area of a square is 5.29 cm^2 . The area of 7 such squares taking into account the significant figures is:

- (1) 37 cm^2 (2) 37.030 cm^2
 (3) 37.03 cm^2 (4) 37.0 cm^2

9 April Evening

Q6

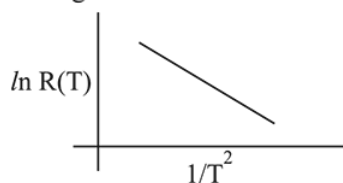
The position vector of a particle changes with time according to the relation $\vec{r}(t) = 15t^2\hat{i} + (4 - 20t^2)\hat{j}$. What is the magnitude of the acceleration at $t = 1$?

- (1) 40 (2) 25 (3) 100 (4) 50

9 April Evening

Q7

In an experiment, the resistance of a material is plotted as a function of temperature (in some range). As shown in the figure, it is a straight line.



One may conclude that:

$$(1) R(T) = \frac{R_0}{T^2} \quad (2) R(T) = R_0 e^{-T_0^2/T^2}$$

$$(3) R(T) = R_0 e^{-T^2/T_0^2} \quad (4) R(T) = R_0 e^{T^2/T_0^2}$$

10 April Morning

Q8

The number density of molecules of a gas depends on their distance r from the origin as, $n(r) = n_0 e^{-ar^4}$. Then the total number of molecules is proportional to :

$$(1) n_0 \alpha^{-3/4} \quad (2) \sqrt{n_0} \alpha^{1/2}$$

$$(3) n_0 \alpha^{1/4} \quad (4) n_0 \alpha^{-3}$$

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JEE Mains 2019 Chapter wise Question Bank

Mathematics in Physics - Answers

Q1

(3) From figure,

$$\vec{r}_G = \frac{a}{2}\hat{i} + \frac{a}{2}\hat{k}$$

$$\vec{r}_H = \frac{a}{2}\hat{j} + \frac{a}{2}\hat{k}$$

$$\therefore \vec{r}_H - \vec{r}_G = \left(\frac{a}{2}\hat{j} + \frac{a}{2}\hat{k}\right) - \left(\frac{a}{2}\hat{i} + \frac{a}{2}\hat{k}\right) = \frac{a}{2}(\hat{j} - \hat{i})$$

10 Jan Morning

Q2

(1) Using, $R^2 = P^2 + Q^2 + 2PQ\cos\theta$

$$4F^2 + 9F^2 + 12F^2\cos\theta = R^2$$

When forces Q is doubled,

$$4F^2 + 36F^2 + 24F^2\cos\theta = 4R^2$$

$$4F^2 + 36F^2 + 24F^2\cos\theta$$

$$= 4(13F^2 + 12F^2\cos\theta) = 52F^2 + 48F^2\cos\theta$$

$$\therefore \cos\theta = -\frac{12F^2}{24F^2} = -\frac{1}{2} \Rightarrow \theta = 120^\circ$$

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Q3

(1) Let magnitude of two vectors $\vec{A}$ and $\vec{B} = a$

$$|\vec{A} + \vec{B}| = \sqrt{a^2 + a^2 + 2a^2\cos\theta} \text{ and}$$

$$|\vec{A} - \vec{B}| = \sqrt{a^2 + a^2 - 2a^2[\cos(180^\circ - \theta)]}$$

$$= \sqrt{a^2 + a^2 - 2a^2\cos\theta}$$

and according to question,

$$|\vec{A} + \vec{B}| = n|\vec{A} - \vec{B}|$$

$$\text{or, } \frac{a^2 + a^2 + 2a^2\cos\theta}{a^2 + a^2 - 2a^2\cos\theta} = n^2$$

$$\Rightarrow \frac{a^2(1+1+2\cos\theta)}{a^2(1+1-2\cos\theta)} = n^2$$

$$\Rightarrow \frac{(1+\cos\theta)}{(1-\cos\theta)} = n^2$$

using componendo and dividendo theorem, we get

$$\theta = \cos^{-1}\left(\frac{n^2-1}{n^2+1}\right)$$

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Q4

(4) Using,

$$R^2 = A_1^2 + A_2^2 + 2A_1A_2\cos\theta$$

$$5^2 = 3^2 + 5^2 + 2 \times 3 \times 5 \cos\theta$$

$$\text{or } \cos\theta = -0.3$$

$$\left(\vec{2A_1} + \vec{3A_2}\right) \cdot \left(\vec{3A_1} - \vec{2A_2}\right) = 2A_1 \times 3A_1$$

$$+ (3A_2)(3A_1)\cos\theta - (2A_1)(2A_2)\cos\theta - 3A_2 \times 2A_2$$

$$= 6A_1^2 + 9A_1A_2\cos\theta - 4A_1A_2\cos\theta - 6A_2^2$$

$$= 6A_1^2 - 6A_2^2 + 5A_1A_2\cos\theta$$

$$= 6 \times 3^2 - 6 \times 5^2 + 5 \times 3 \times 5(-0.3)$$

$$= -118.5$$

8 April Evening

Q5

$$(4) A = 7 \times 5.29 = 37.03 \text{ cm}^2$$

The result should have three significant figures, so

$$A = 37.0 \text{ cm}^2$$

9 April Evening

Q6

$$(4) \vec{r} = 15t^2\hat{i} + (4 - 20t^2)\hat{j}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = 30t\hat{i} - 40t\hat{j}$$

$$\text{Acceleration, } \vec{a} = \frac{d\vec{v}}{dt} = 30\hat{i} - 40\hat{j}$$

$$\therefore a = \sqrt{30^2 + 40^2} = 50 \text{ m/s}^2$$

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Q7

(2) Equation of straight line from graph

$$y = -mx + c$$

$$\Rightarrow \ln R = -m\left(\frac{1}{T^2}\right) + c$$

here, m & c are constants

$$R = e^{-m\left(\frac{1}{T^2}\right) + c}$$

$$= e^{-m\left(\frac{1}{T^2}\right)} \times e^c$$

$$R(T) = R_0 e^{\frac{-T_0^2}{T^2}}$$

10 April Morning

Q8

$$(1) N = \int \rho(dv)$$

$$= \int_0^r n_0 e^{-\alpha r^4} \times 4\pi r^2 dr = 4\pi n_0 \int_0^r r^2 (e^{-\alpha r^4}) dr$$

$$\propto n_0 \alpha^{-3/4}$$

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